Lexington Whalen

linkedin.com/in/lxaw

Summary

- **Machine Learning Research Leadership:** Lead inference optimization research at SoftBank's SB Intuitions, pioneering the organization's first-ever speculative decoding implementation and leading algorithmic inference optimizations for Japan's largest LLM series. Co-led the development of flagship diffusion language models at NVIDIA, designing novel training and inference infrastructure for models of 1B to 14B parameters. Team scaled from 2 people to multiple teams. Work was presented directly to CEO Jensen Huang and training/inference recipes used by 5+ internal research teams. Design led to ∼10x higher speed and ∼10% improvements in accuracy as compared to leading models at the time.
- **Research Impact:** Co-led some of the first diffusion language models at NVIDIA, building the first training and inference framework used by 5+ research teams. Led interdisciplinary teams of Ph.D. and Masters students in pioneering diffusion model research, achieving 5x training efficiency improvements while preserving model quality. Published at premier computer vision conferences, with research methodologies adopted by NVIDIA for internal development. Demonstrated expertise in optimizing training and inference strategies for both state-space and transformer architectures.
- **Machine Learning Engineering:** Developed comprehensive evaluation and training pipelines for NVIDIA's internal diffusion-LLM projects, scaling model training from millions to billions of parameters across billions to trillions of tokens. Experience with distributed training systems leveraging thousands of GPUs with advanced parallelization strategies.
- **Healthcare AI:** Technical Lead on United States' THOR healthcare initiative, developing innovative tiny PyTorch models that reduced cancer detection sensor degradation by over 70%. Collaborate with researchers at top institutions like Carnegie Mellon University, Stanford, Purdue, Georgia Tech, et cetera.
- **Recognition:** NSF Graduate Research Fellowship recipient ($150,000 value), University of South Carolina Top Scholar Award ($100,000+ value), and Critical Language Scholar of Japanese (U.S. Department of State, only 500 selected among 5,000+ applicants).
- **Technical Development:** Full-stack developer for research study websites using Django/React/SQL, managing databases with thousands of participant entries. Developed and won grants totaling $20,000+, leading teams of developers.

Experience

SoftBank – SB Intuitions — **Tokyo, Japan**
R&D Engineer, Leadership & Professional Track (L/P Grade 3) — *Apr. 2026 –*

- Co-lead pretraining and post-training optimization research for the Sarashina series of large language models at SoftBank's SB Intuitions R&D Division.
- Led SB Intuitions' first-ever inference optimizations using speculative decoding, delivering significant throughput improvements for the Sarashina LLM series.
- Led algorithmic optimizations for model inference, improving serving efficiency and reducing latency across the Sarashina model family.
- Serve as a key technical voice in Japanese-language cross-team discussions on pretraining, post-training, and inference optimization, bridging research and engineering across internal teams.
- Collaborate with leading Japanese and international technology companies to evaluate GPU resources and co-develop long-term compute strategies for large-scale model training.
- Selected as SoftBank Group Representative for the 2026 Entrance Ceremony, chosen to represent the incoming cohort across SoftBank Group companies.

NVIDIA — **Atlanta, GA & Santa Clara, CA**
Efficient Deep Learning Intern — *Mar. 2025 – Mar. 2026*

- Trained, finetuned, and evaluated the flagship series of NV Research 1–14B diffusion language models across base, instruct, and RL variants. Built the evaluation and inference infrastructure, adopted by 5+ internal research teams for vision-language models, embeddings, and latent reasoning, with over 1 million internal downloads.
- Worked on model distillation, efficient training recipes, and efficient inference recipes.
- Specialized in diffusion models on high-performance computing infrastructure utilizing clusters of thousands of NVIDIA GPUs on SLURM-managed servers.
- Presented weekly progress reports and technical findings to audiences of 20+ senior researchers and cross-functional engineering teams.
- Developed EfficientDLM series of diffusion language models that outperform Qwen3 autoregressive series in both accuracy and speed. Over 1M internal downloads.
- Worked with vision team to release Nemotron series of diffusion language models; expected release in March of 2026.

NVIDIA **Atlanta, GA**
Data Filtering Challenge Lead *Dec. 2024 – Dec. 2025*

- Led development and management of challenge website utilizing Google Analytics to track engagement and optimize user experience across 30+ participating university research teams and companies.
- Assisted in profiling of initial 400M parameter baseline model for participants, creating standardized benchmarks to evaluate submission quality.
- Designed and implemented a comprehensive evaluation framework leveraging 10B token fine-tuning datasets to ensure consistent, fair assessment of model submissions.
- Promoted challenge at International Conference on Machine Learning, ranked in the top 3 machine learning conferences globally with ∼9,000 attendees and <25% paper acceptance rate.
- Coordinated with corporate sponsors including Lambda Labs and Turing to secure GPU resources, evaluating team submissions and managing the end-to-end competition workflow.

Georgia Institute of Technology **Atlanta, GA**
Machine Learning Engineer *Aug. 2024 – Dec. 2025*

- Led team of 3 Ph.D. and Masters students to design innovative methods of reducing diffusion model training time by up to 5x on average while maintaining generation quality. Method was accepted to Computer Vision and Pattern Recognition (CVPR) 2025.
- Algorithm lead for United States' ARPA-H THOR healthcare initiative. Spearheaded effort to reduce cancer detection sensor degradation by over 70%, improving the lifetime of the sensors from only a few hours to several days (over a 10x improvement).
- Engineered resource-efficient machine learning architectures designed to be powered by energy from the human body.
- Collaborated with 5+ teams from top universities like Stanford, Northwestern, Carnegie Mellon, and MIT, representing Georgia Tech.
- Traveled nationally to present technical progress to federal sponsors and healthcare stakeholders, effectively communicating complex technical concepts to diverse audiences.

University of South Carolina **Columbia, SC**
Software Developer *Jan. 2021 – Aug. 2024*

- Designed and built three full-stack research study websites using PHP, Django, React, and SQL.
- Architected and managed SQL databases of thousands of participant entries, ensuring data integrity, security compliance, and optimized query performance.
- Led the development of competitive federal and state grant applications, winning $20,000+ in funding.

- Led teams of 4+ developers, providing mentorship and technical guidance while meeting KPIs and performance targets.
- Implemented advanced machine learning techniques including clustering algorithms, random forests, and neural networks to analyze patient data, identify patterns, and develop predictive models.
- Developed an innovative automated language similarity analysis system using numpy and pandas that reduced document comparison time by approximately 99% (from 2–3 days to under 10 minutes). Published in Linguistic Society of America.

Selected Publications & Patents

- Y. Fu, **L. Whalen**, X. Dong, S. Diao, C. Wu, E. Xie, S. Han, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Training Framework for Converting Autoregressive Language Models into Faster Diffusion Language Models." *U.S. Patent, to be filed*, 2026. [NVIDIA]
- C. Shih, C. Li, C. Yu, H. Fang, S. Li, W. Hsin, **L. Whalen**, H. Suh, G. Eisenhauer, L. Liu, Y. (Celine) Lin. "NeRArch-Sim: A Unified Simulator for Benchmarking and DSE of Neural Rendering Accelerators." *International Symposium on Computer Architecture (ISCA)*, 2026. [Georgia Tech]
- Y. Fu*, **L. Whalen***, Z. Ye, X. Dong, S. Diao, J. Liu, C. Wu, H. Zhang, E. Xie, S. Han, M. Khadkevich, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Efficient-DLM: From Autoregressive to Diffusion Language Models, and Beyond in Speed." *International Conference on Machine Learning (ICML)*, 2026. [NVIDIA]
- **L. Whalen***, Z. Du*, H. You*, C. Li, S. Li, Y. (Celine) Lin. "Early-Bird Diffusion: Investigating and Leveraging Timestep-Aware Early-Bird Tickets in Diffusion Models for Efficient Training." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [Georgia Tech]
- Z. Ye, Z. Wang, K. Xia, J. Hong, L. Li, **L. A. Whalen**, C. Wan, H. You, C. Lin, S. Kundu. "LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering." *Association for Computational Linguistics (ACL)*, 2025. [Georgia Tech]
- A. Murphy, **L. Whalen**, S. Dubinsky, M. Gavin, J. F. Bailyn, J. Ginn. "On 'historical unity' of Russian and Ukrainian: A linguistic perspective on language conflict and change." *Linguistic Society of America (LSA)*, Apr. 2023. [University of South Carolina]

Education

Georgia Institute of Technology — **Atlanta, GA**
Graduate Researcher – Efficient Machine Learning Systems — *Aug. 2024 – Dec. 2025*

University of South Carolina — **Columbia, SC**
Accelerated Master's in Computer Science — *Jan. 2024 – Aug. 2024*

University of South Carolina — **Columbia, SC**
Bachelor's in Computer Science — *Jan. 2021 – Dec. 2023*

Awards & Recognition

- SoftBank Group Representative for Entrance Ceremony 2026 — Mar. 2026
- NSF GRFP Awardee ($150,000 value) — Apr. 2024
- Toyo University Exchange Student Representative — Jul. 2023
- University of South Carolina Outstanding Senior, Class of 2023 — Feb. 2023
- U.S. Department of State Critical Language Scholar of Japanese — Jan. 2023
- University of South Carolina Top Scholar ($100,000+ value) — Aug. 2020

Qualifications

- Japanese Language Proficiency Test (N1)

Language Skills

- English – Native Level

- Japanese – Native Level (N1 Certified)
- Mandarin – Intermediate Level
- German – Beginner